

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Owned end-to-end design and development of **cross-platform system integrating** wearable data for 100+ users, enabling scalable **data ingestion** and analytics

• Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%

• Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by **43%**

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by **41%**

• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection

• Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Built and deployed scalable, microservices-based Retrieval-Augmented Generation (**RAG**) system using **FastAPI**, **PostgreSQL**, and vector embeddings

• Designed efficient, fault-tolerant **ETL** pipelines to process and index large document datasets for low-latency retrieval

• Improved system performance by **64%** through optimized indexing strategies and pipeline efficiency improvements

• Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive dashboards in Power BI and Looker Studio, segmenting performance data by user cohort and source, reducing reporting time by 40% and improving operational visibility into KPIs

• Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure patterns, trends, and performance issues

• Developed SQL queries to audit data flows, validate data integrity, and detect tracking gaps and quality issues across operational datasets, accelerating root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained 20+ dashboards in Power BI and Looker Studio across Sales, Finance, and Operations, using SQL to track KPIs, funnel metrics, and deliver stakeholder performance reports

• Automated recurring **Excel**-based reporting workflows using **VBA**, reducing report preparation time by **42%** and improving data consistency across weekly reporting cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of 6, analyzing user activation metrics, funnel performance, and segmented cohort data to guide A/B testing decisions and data-driven feature prioritization

• Tracked delivery metrics and QA issues through JIRA workflows, documenting data definitions, metric calculations, and preparing weekly performance summaries for stakeholder review

PROJECTS

FMCG Sales Demand Forecasting

March 2025

View Project

• Analyzed FMCG **sales demand data** across categories and locations to identify trends, seasonal patterns, and key drivers impacting inventory

• Built and compared ARIMA, SARIMA, and regression models applying statistical hypothesis testing, confidence intervals, and significance analysis to evaluate demand patterns and improve forecast accuracy

• Derived actionable insights on promotions, pricing, and stock behavior, enabling better inventory planning and reducing stockouts and overstocking risks